

Measurement of Absolute Single Differential Cross-Section in p_T Bins using Runs 2 and 3 Data (Addressing comments received for DocDB 11535)

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SeaQuest Experiment (E906)

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- Events from runs 2 and 3
- Kinematic Phase Space
- Event Selection Criteria
- Comments received on previous talk:
 - $d\sigma/dp_T^2$ Distribution for different x_F bins
 - $d\sigma/dp_T^2$ Distribution and ratio plot
 - Cross-section ratio as a function of x_F
- Cross-section ratio from different roadsets to RS67
- Next Steps

Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:

/sequest/users/harshaka/e906.project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2.1_1138.root
- merged_RS57_LD2.3_1138.root
- merged_RS57_Empty.2_1138.root
- merged_RS59_LH2.1_465.root
- merged_RS59_LD2.3_466.root
- merged_RS59_Empty.2_466.root
- merged_RS62_LH2.1_1234.root
- merged_RS62_LD2.3_1237.root
- merged_RS62_Empty.2_1234.root
- merged_RS70_LH2.1_264.root
- merged_RS70_LD2.3_266.root
- merged_RS70_Empty.2_267.root

Abi's ROOT files:

/sequest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.8]
- **p_T Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- **x_F Bins:** [0.0, 0.80] with a fixed step width of 0.05.

Event Selection Criteria (Chuck Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi^2_{\text{target}} < 15$, $\chi^2_{\text{dimuon}} < 18$, and $\chi^2/(\text{nHits} - 5) < 12$.
- Hit requirements: $\text{nHits} > 13$ per track, $\text{nHits}_1 + \text{nHits}_2 > 29$, St1 total > 8 .
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.
- $20 < D1 < 385$ (global reconstruction efficiency curve defined in this range)

ONLY RS62, $\text{runID} > 11500$ applied to ensure correct target positions of dimuons

Note: Currently using the same set of Chuck Cuts defined in DocDB 2111.V42

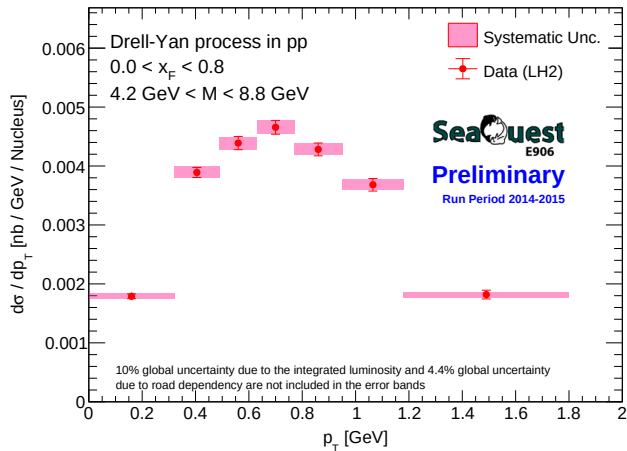
POT Values Used

Roadset	LH2	LD2	Flask
57	3.533324×10^{16}	1.768358×10^{16}	3.918550×10^{15}
59	9.365986×10^{15}	4.319952×10^{15}	1.010350×10^{15}
62	5.281659×10^{16}	2.382899×10^{16}	1.097456×10^{16}
67	1.611435×10^{17}	7.694541×10^{16}	3.662417×10^{16}
70	1.785745×10^{16}	8.752588×10^{15}	3.841280×10^{15}

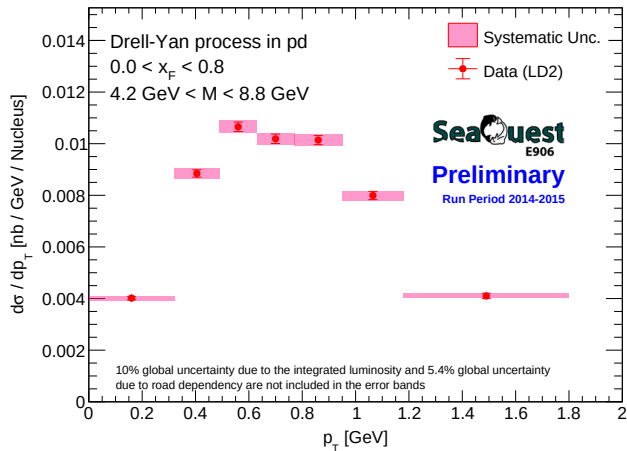
Note: POT Values from runs mixed by Harsha in DocDB 11524.

- Generate single differential cross-section as a function of p_T^2
- Calculate cross-section ratio
- Generate single differential cross-section as a function of p_T for different x_F bins
- Calculate cross-section ratio
- Generate cross-section ratio as a function of different x_F bins

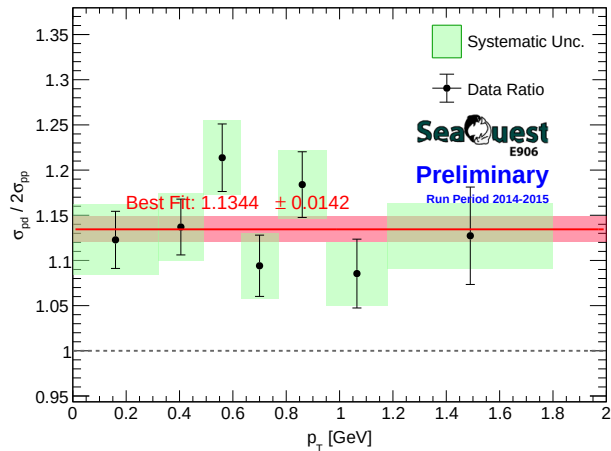
CrossSection LH2 geom vs pT with logo



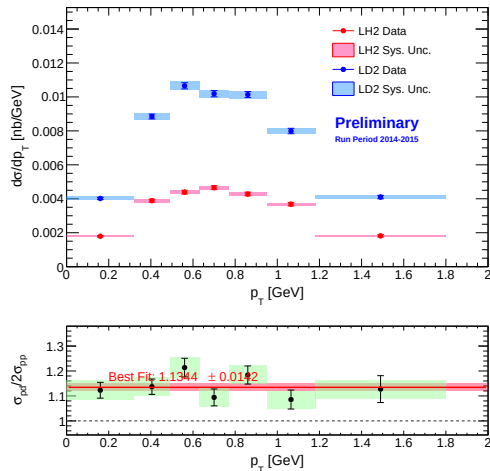
CrossSection LD2 geom vs pT with logo



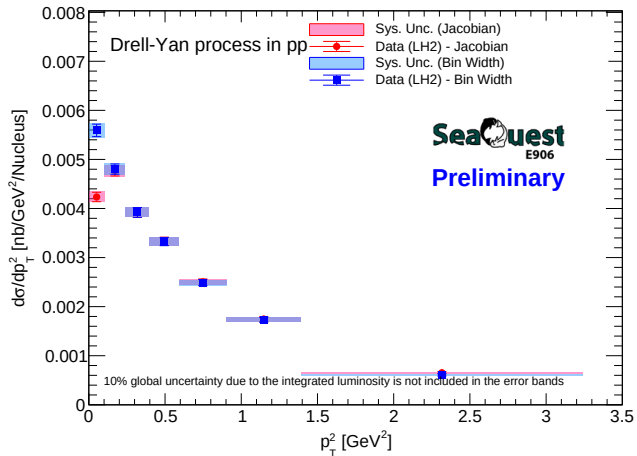
CrossSection Ratio pd 2pp vs pT geom with logo



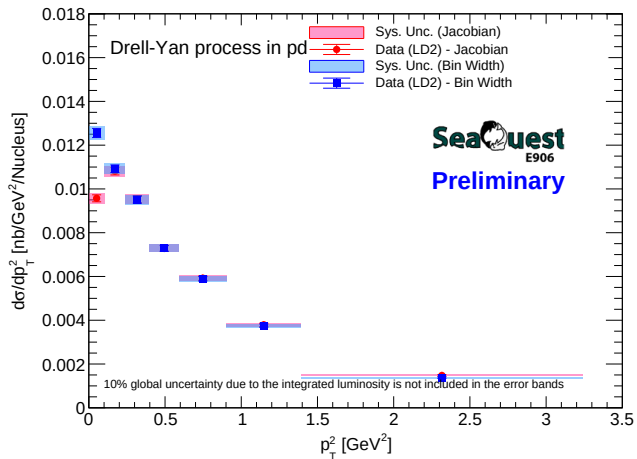
Combined XSec Ratio vs pT geom



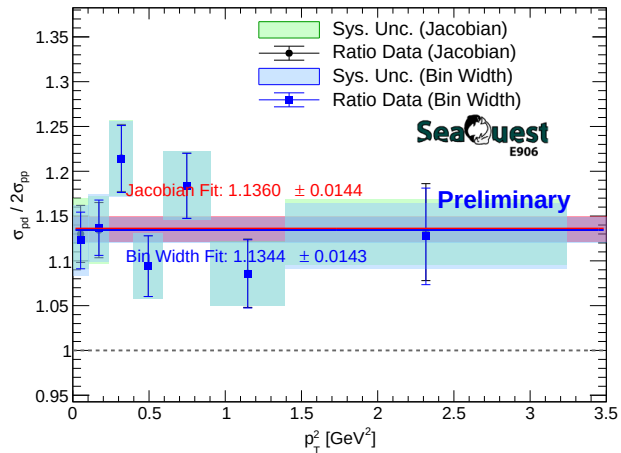
CrossSection LH2 geom pT2 with logo



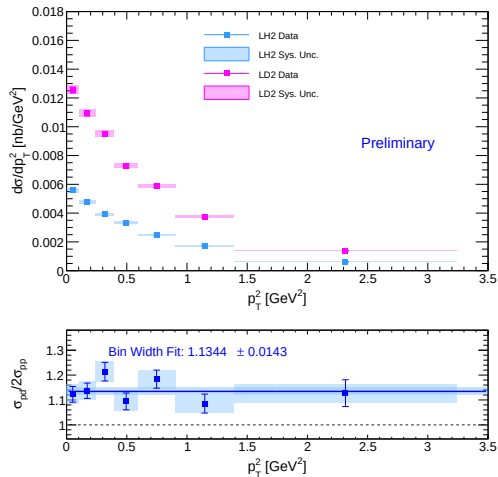
CrossSection LD2 geom pT2 with logo



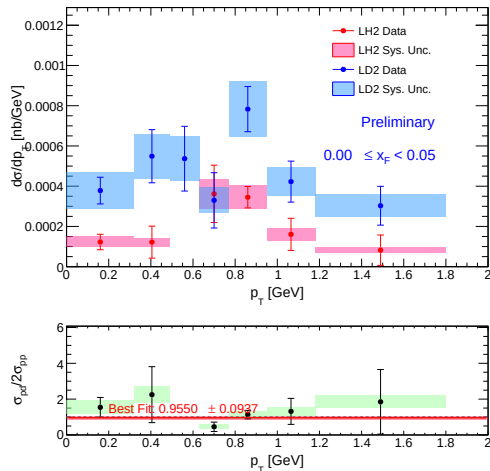
CrossSection Ratio pd 2pp geom pT2 with logo



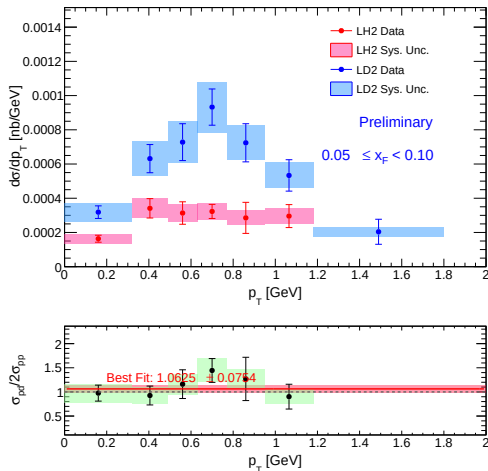
Combined XSec Ratio geom pT2



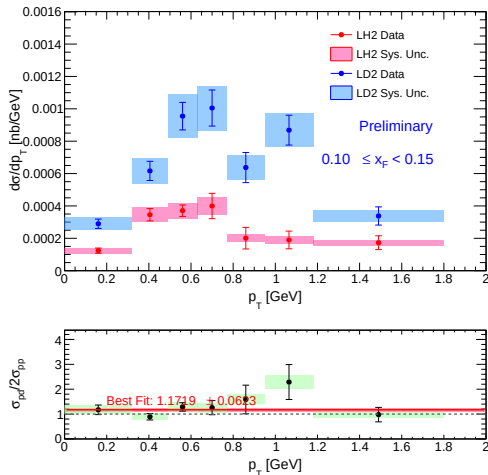
Single Differential Cross-Sections $0.00 < x_F < 0.05$



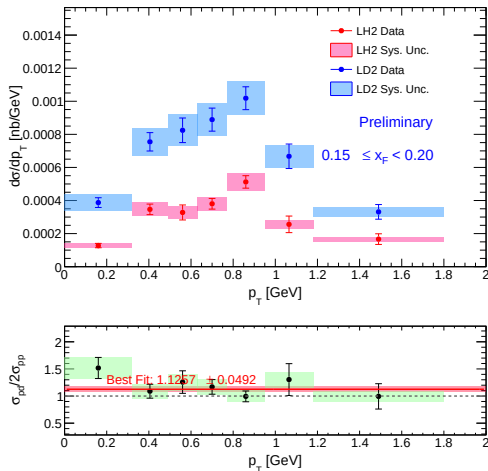
Single Differential Cross-Sections $0.05 < x_F < 0.10$



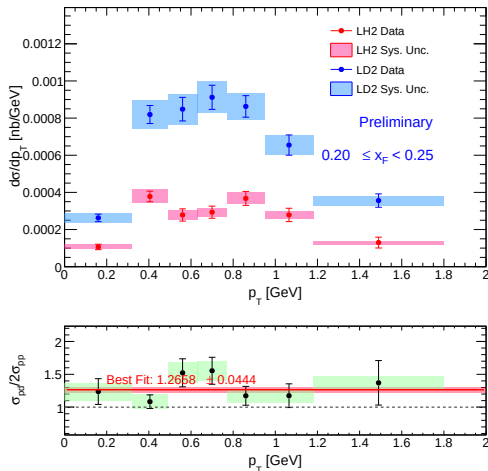
Single Differential Cross-Sections $0.10 < x_F < 0.15$



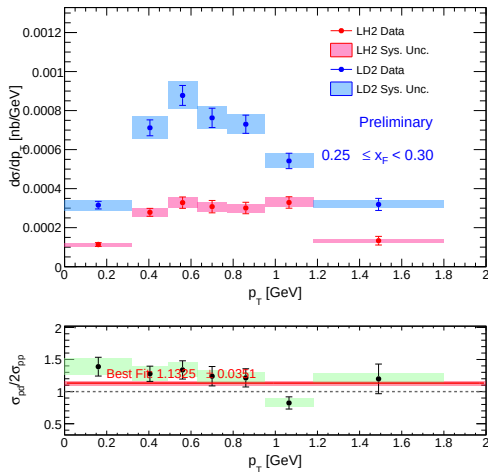
Single Differential Cross-Sections $0.15 < x_F < 0.20$



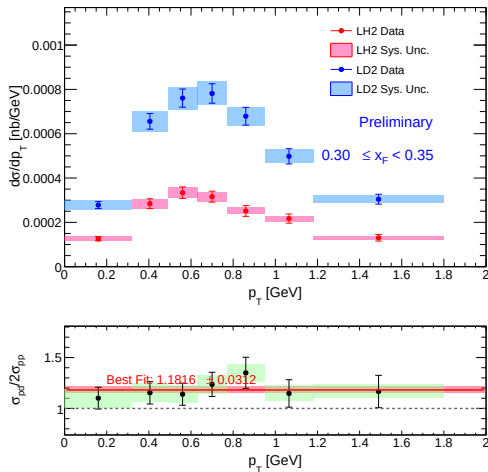
Single Differential Cross-Sections $0.20 < x_F < 0.25$



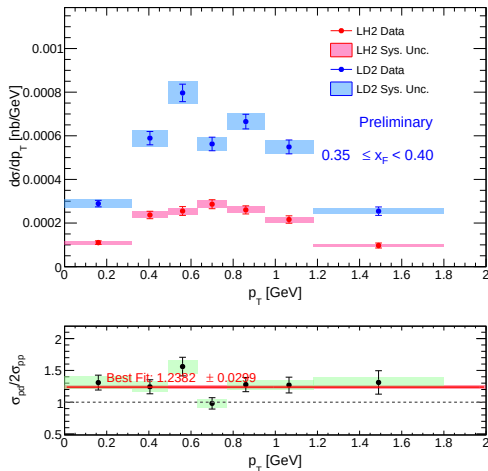
Single Differential Cross-Sections $0.25 < x_F < 0.30$



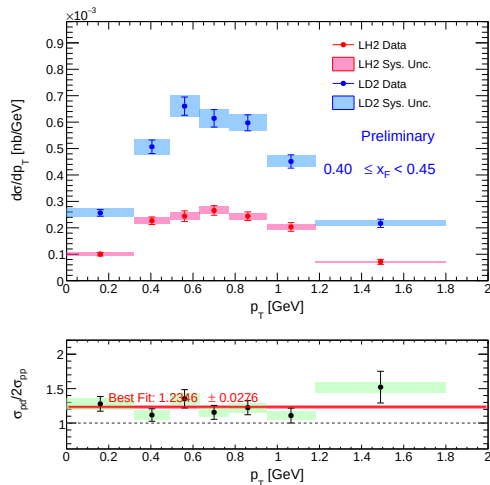
Single Differential Cross-Sections $0.30 < x_F < 0.35$



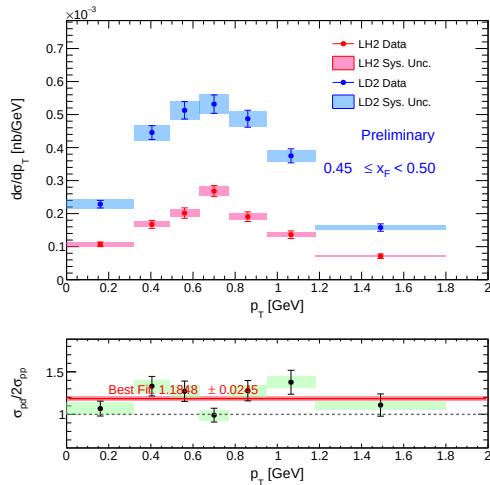
Single Differential Cross-Sections $0.35 < x_F < 0.40$



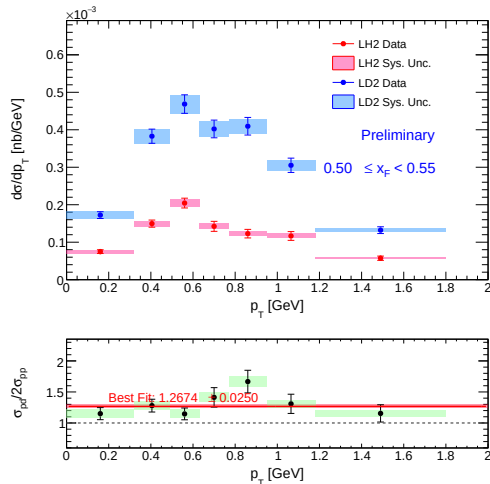
Single Differential Cross-Sections $0.40 < x_F < 0.45$



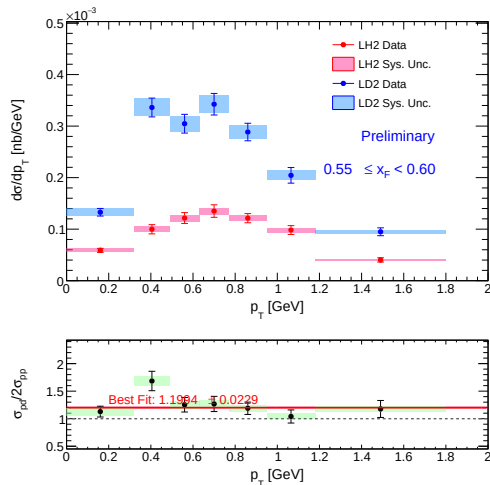
Single Differential Cross-Sections $0.45 < x_F < 0.50$



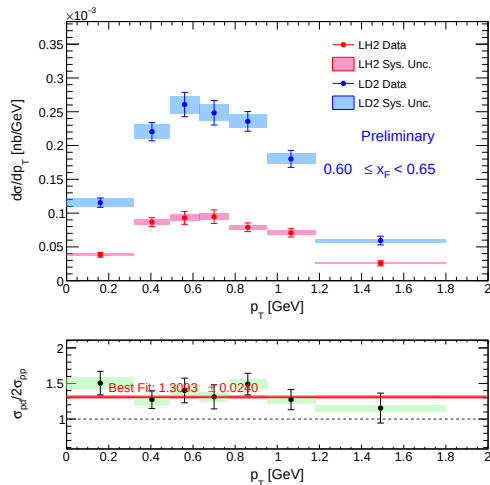
Single Differential Cross-Sections $0.50 < x_F < 0.55$



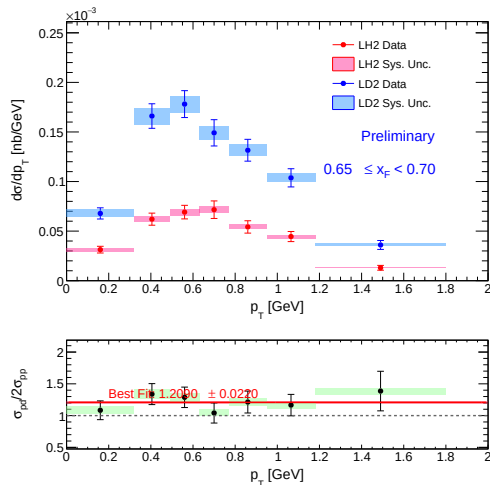
Single Differential Cross-Sections $0.55 < x_F < 0.60$



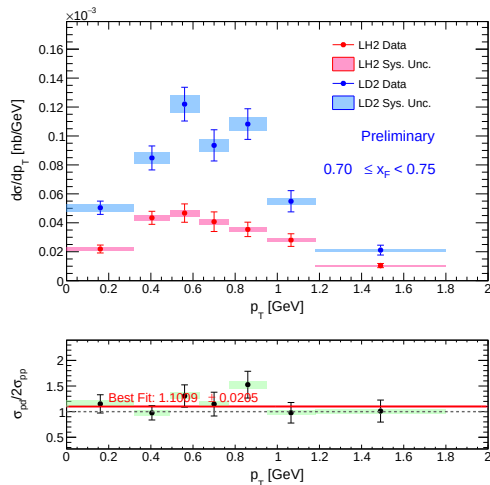
Single Differential Cross-Sections $0.60 < x_F < 0.65$



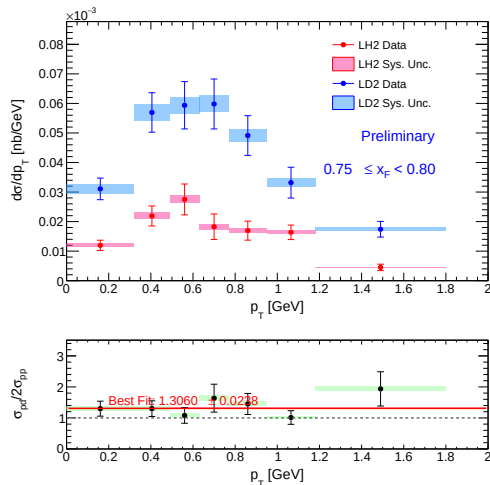
Single Differential Cross-Sections $0.65 < x_F < 0.70$



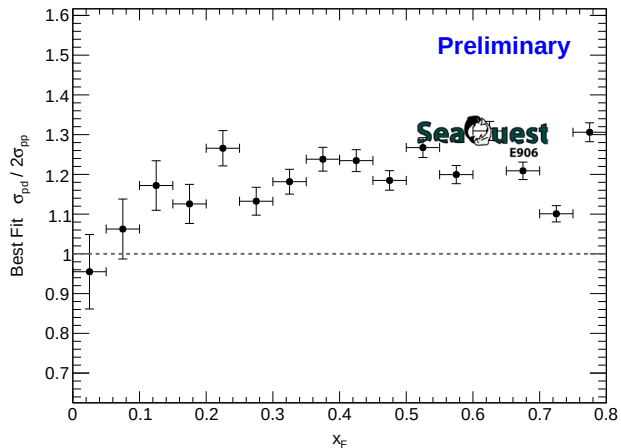
Single Differential Cross-Sections $0.70 < x_F < 0.75$



Single Differential Cross-Sections $0.75 < x_F < 0.80$

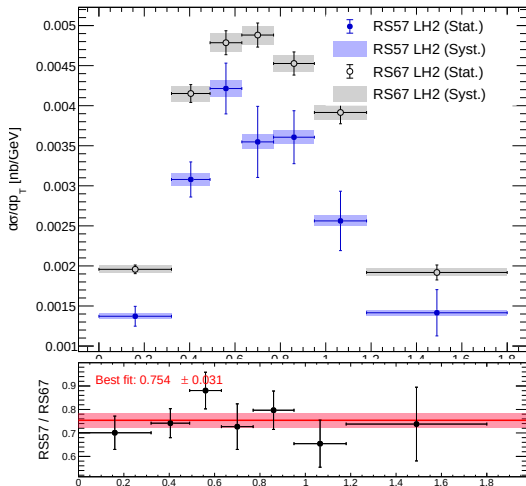


Cross-Section ratio Vs x_F

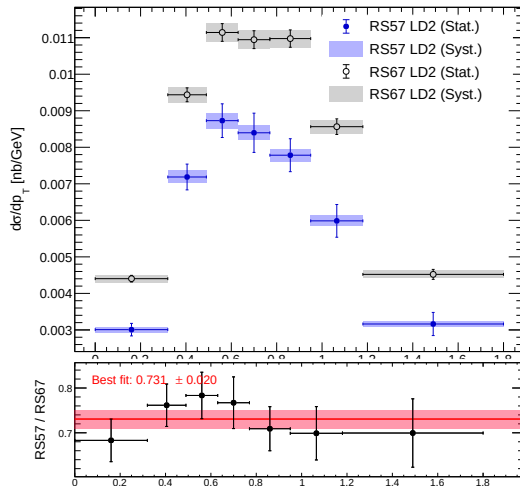


Cross-Section Ratio RS57/RS67

LH2 Target

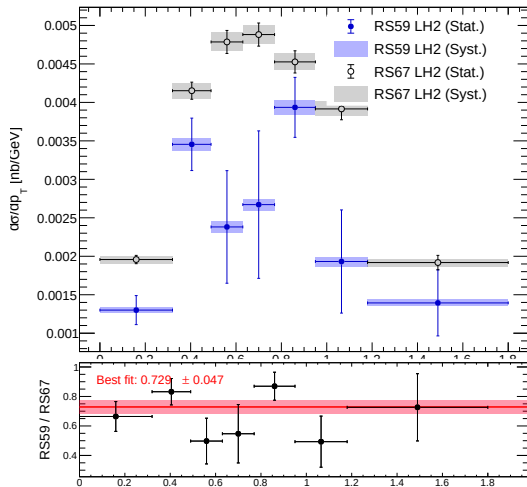


LD2 Target

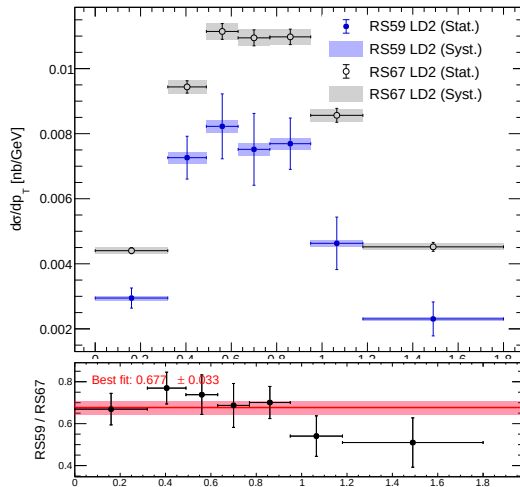


Cross-Section Ratio RS59/RS67

LH2 Target

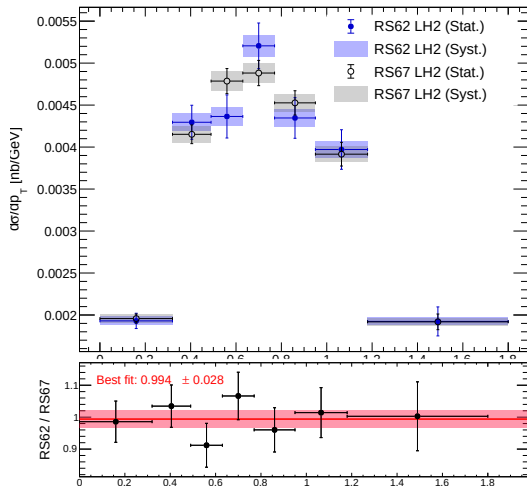


LD2 Target

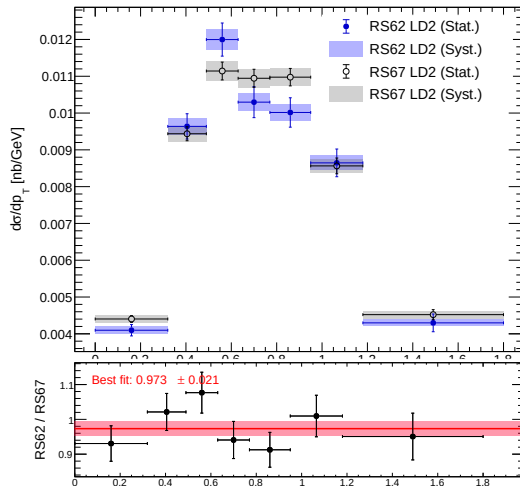


Cross-Section Ratio RS62/RS67

LH2 Target

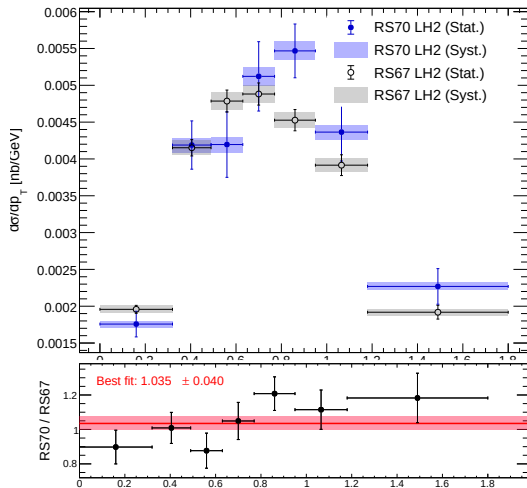


LD2 Target

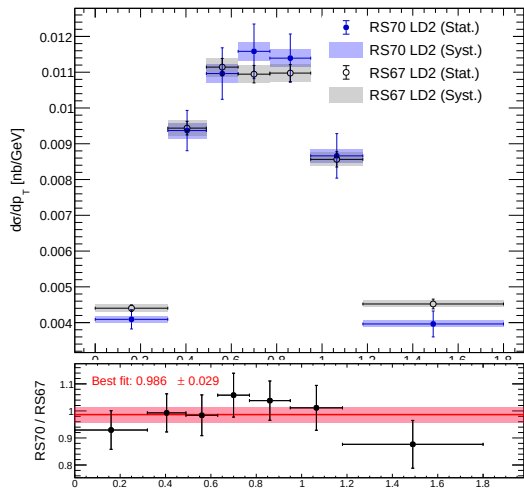


Cross-Section Ratio RS70/RS67

LH2 Target

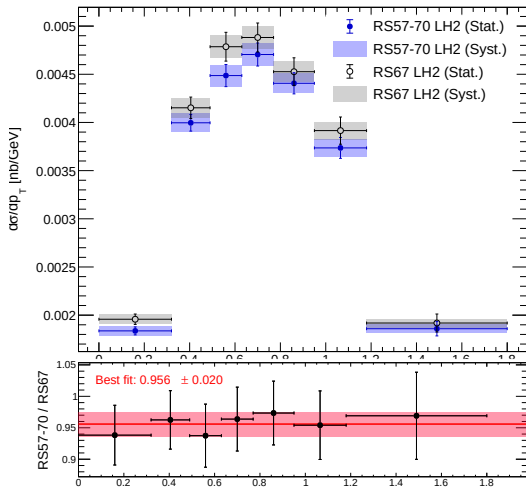


LD2 Target

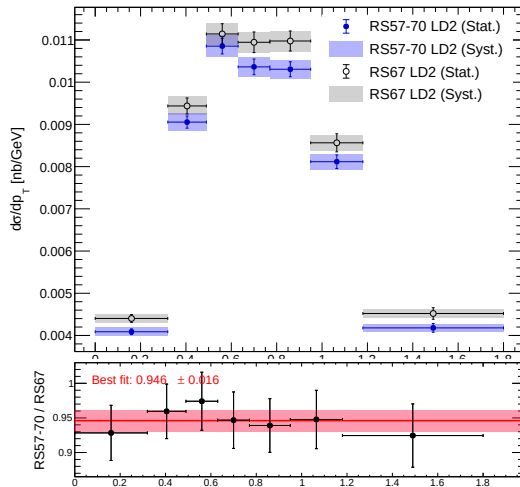


Cross-Section Ratio RS57-70/RS67

LH2 Target



LD2 Target



Summary

Roadset	57	59	62	70	57-70
Cross-Section Ratio: Roadset/RS67 (LH2)	0.754 ± 0.031	0.729 ± 0.047	0.994 ± 0.028	1.035 ± 0.040	0.956 ± 0.020
Cross-Section Ratio: Roadset/RS67 (LD2)	0.731 ± 0.020	0.677 ± 0.033	0.973 ± 0.021	0.986 ± 0.029	0.946 ± 0.016

Note: These cross-section ratios are consistent with Harsha's numbers mentioned in DocDB: 11539

Next Steps

- Need to work on unfolding (work on progress).
- Planning to present preliminary plots during upcoming APS DPF meeting.
- Analysis note will be written for this Run 2-3 release, as a preliminary result.

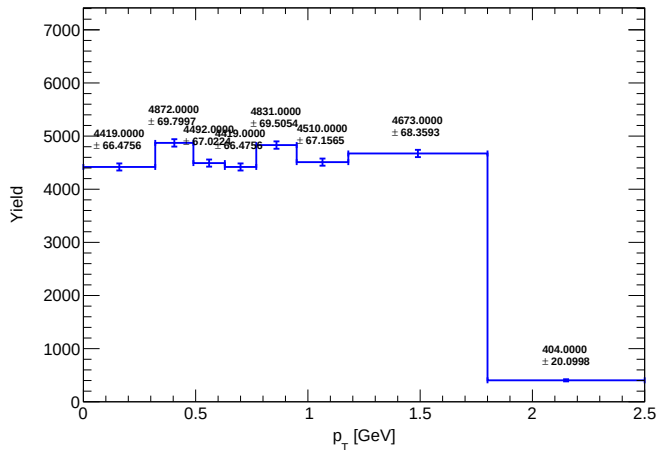
Backup Slides

$$\epsilon_{\text{signal}} = \frac{\langle \epsilon \rangle_{\text{total}} Y_{\text{total}} - \langle \epsilon \rangle_{\text{mix}} Y_{\text{mix}}}{Y_{\text{total}} - Y_{\text{mix}}}$$

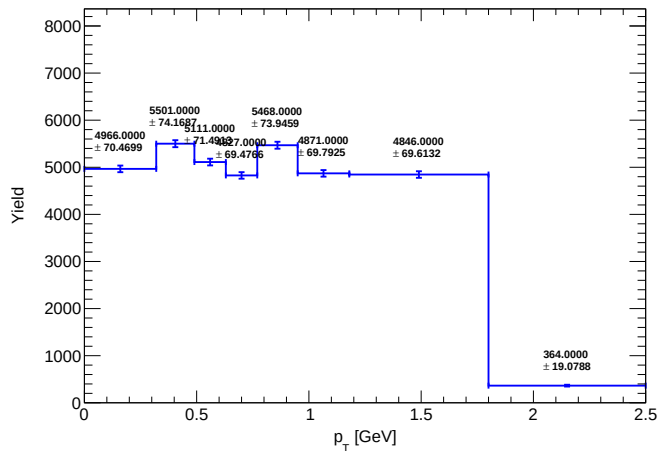
Yield Distributions

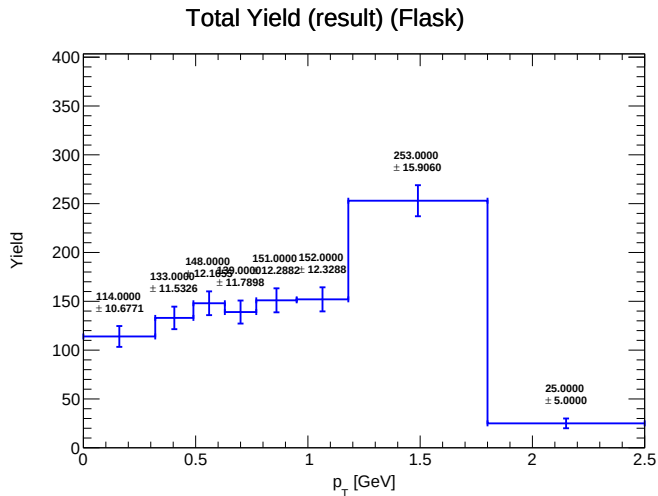
Yield calculations for LH2, LD2, and Empty Flask targets.

Total Yield (result) (LH2)

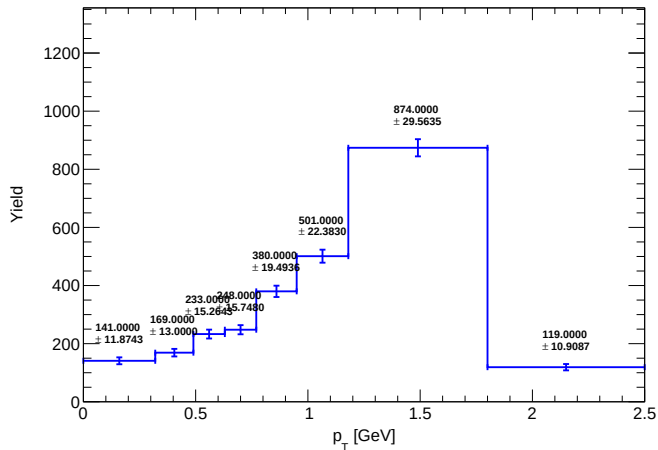


Total Yield (result) (LD2)

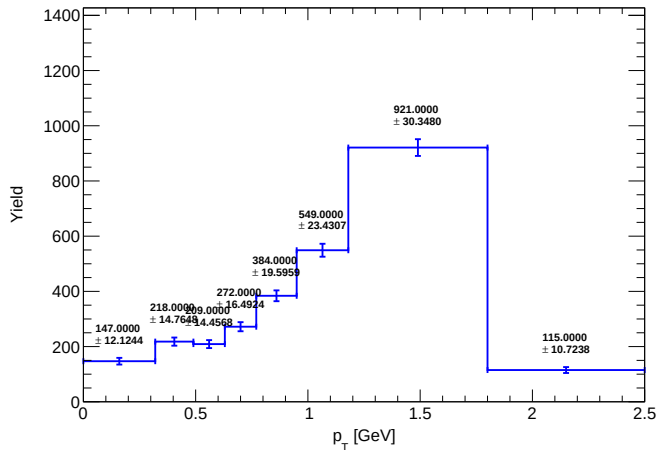




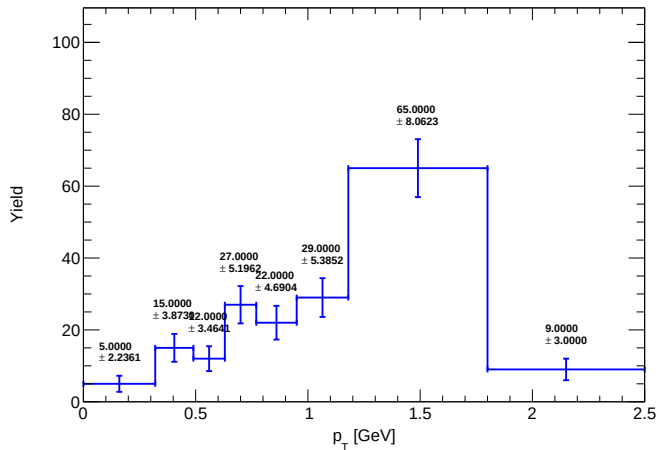
Mix Yield (result_mix) (LH2)



Mix Yield (result_mix) (LD2)

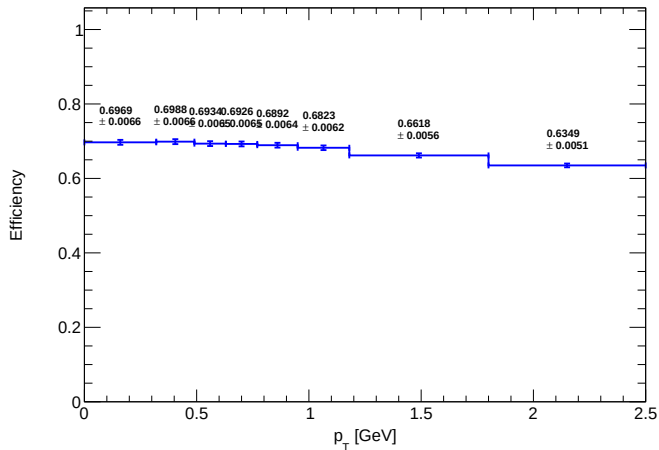


Mix Yield (result_mix) (Flask)

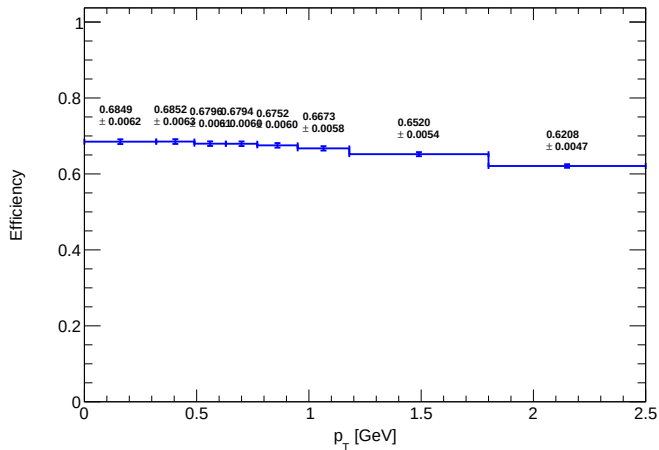


Reco efficiency calculated using D1 occupancy and global efficiency curve.

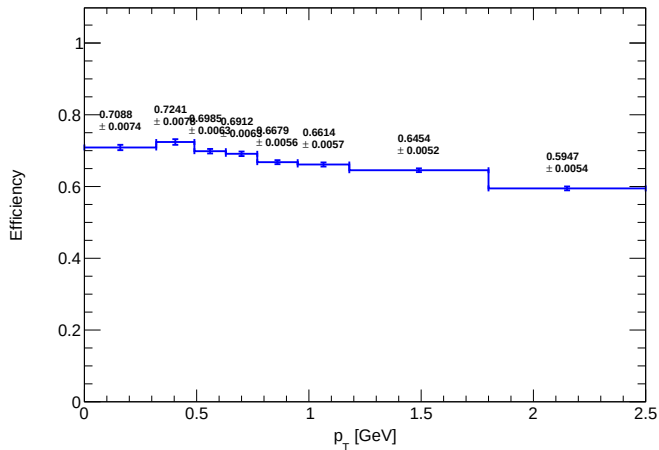
Avg Reco Eff (Total) (LH2)

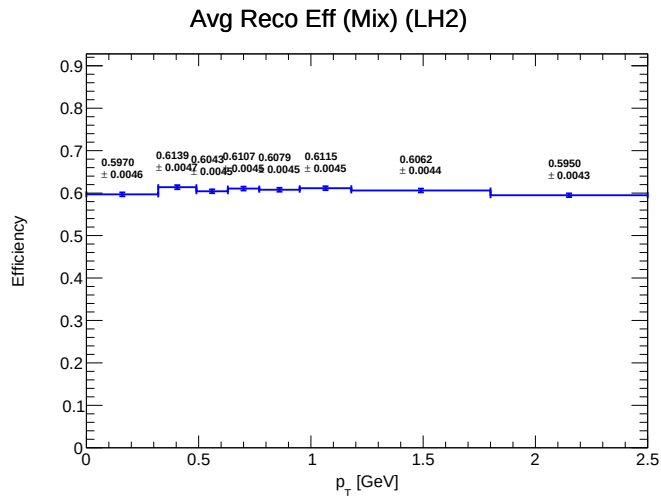


Avg Reco Eff (Total) (LD2)

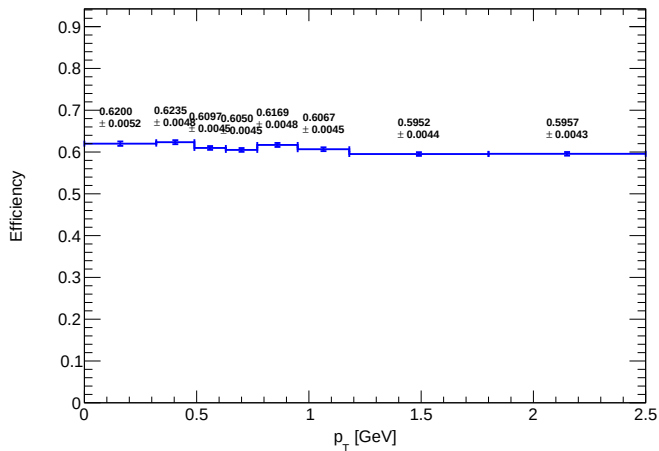


Avg Reco Eff (Total) (Flask)

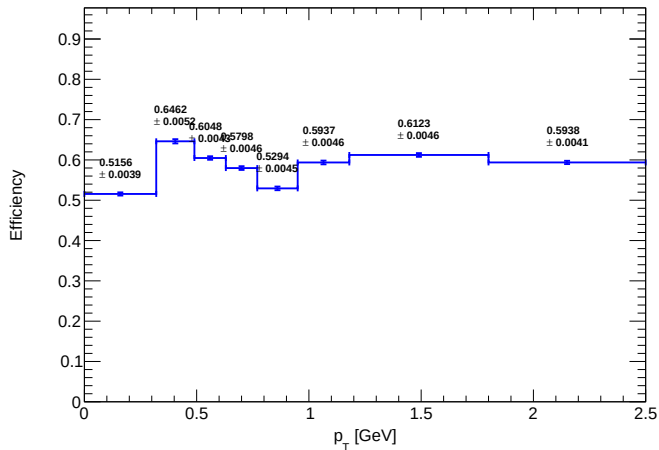




Avg Reco Eff (Mix) (LD2)

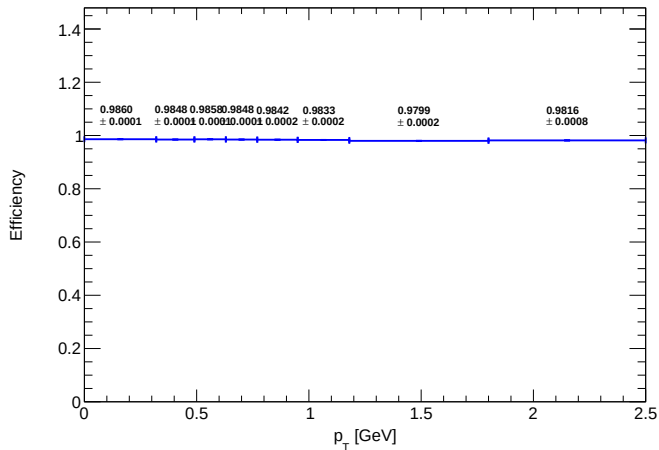


Avg Reco Eff (Mix) (Flask)

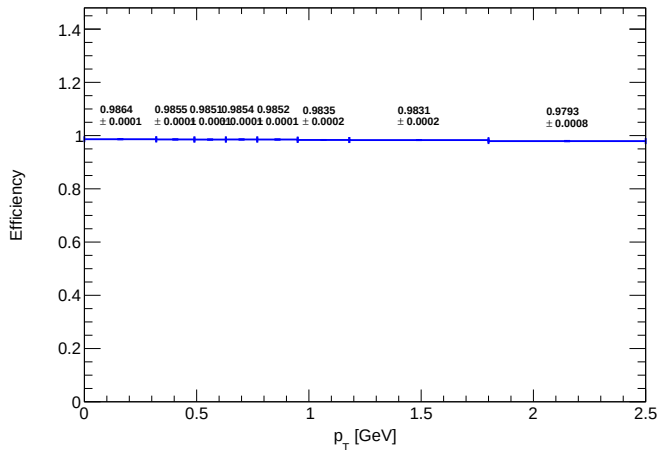


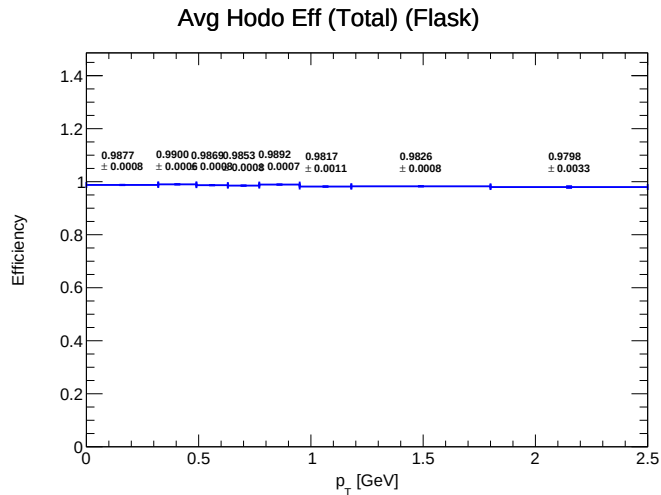
Hodoscope efficiencies propagated from latest paddle efficiency tables (DocDB 11467).

Avg Hodo Eff (Total) (LH2)

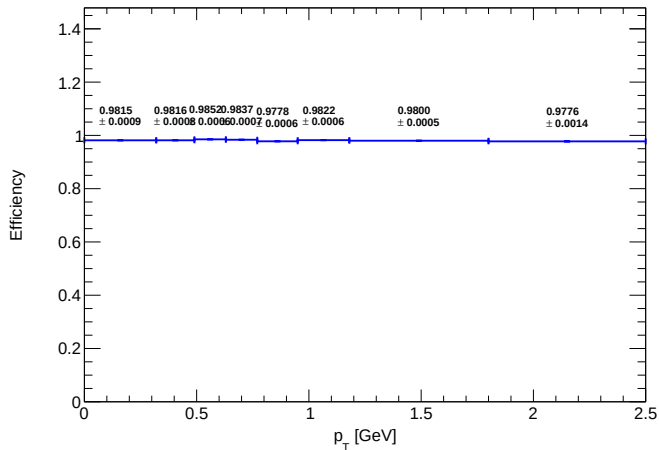


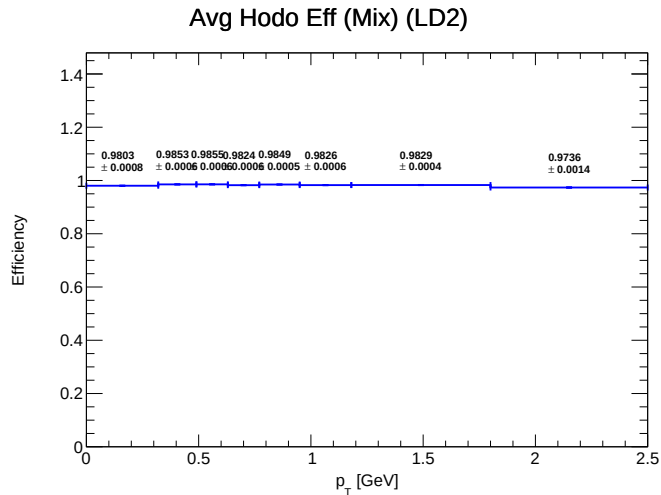
Avg Hodo Eff (Total) (LD2)



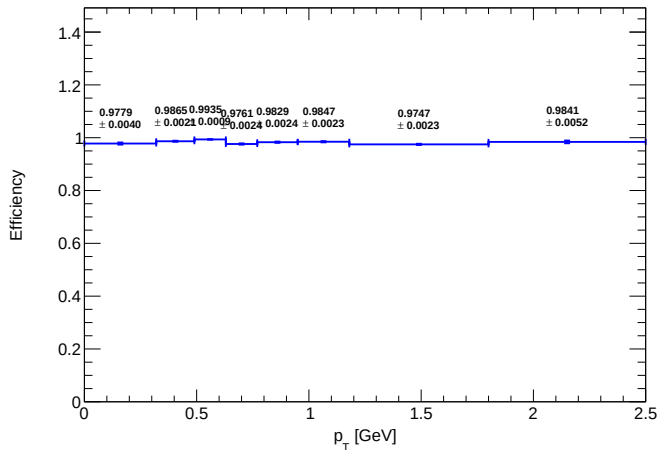


Avg Hodo Eff (Mix) (LH2)

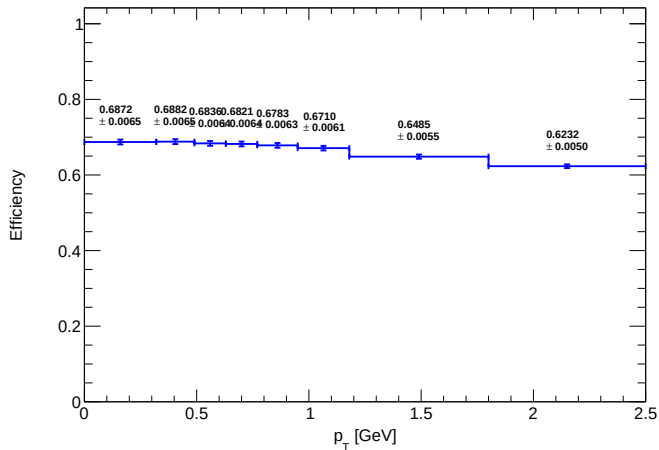




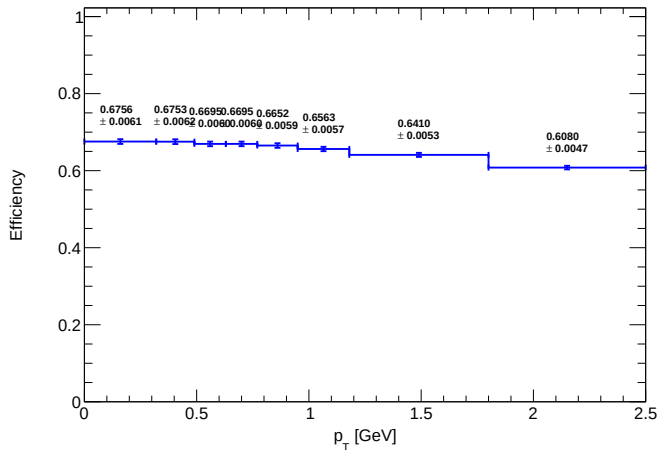
Avg Hodo Eff (Mix) (Flask)



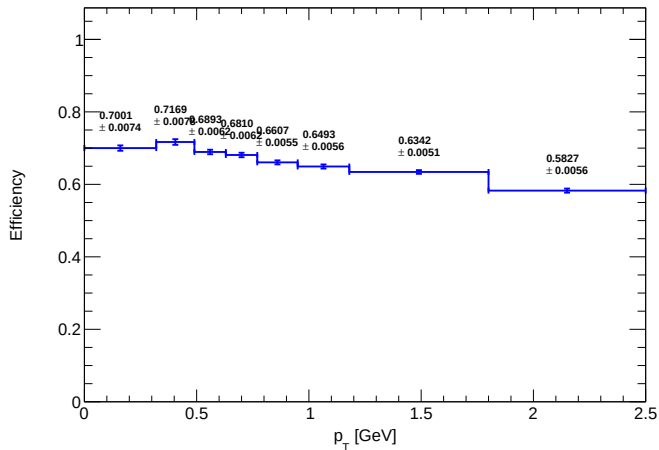
Avg Final Eff (Total) (LH2)

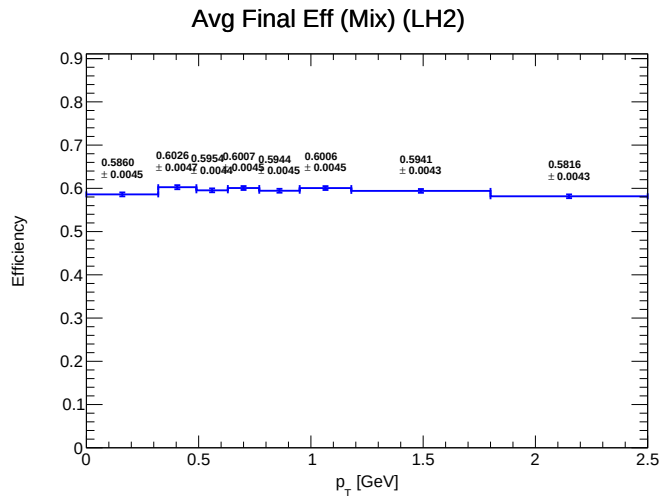


Avg Final Eff (Total) (LD2)

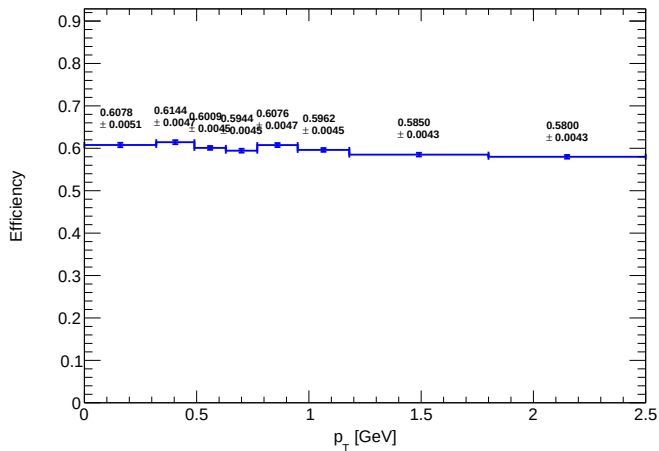


Avg Final Eff (Total) (Flask)

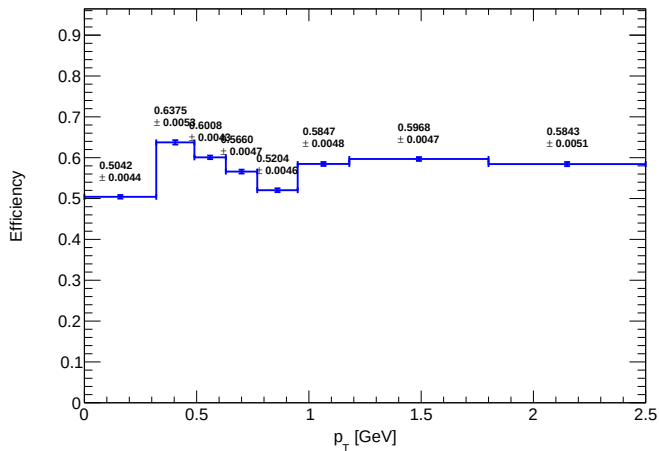




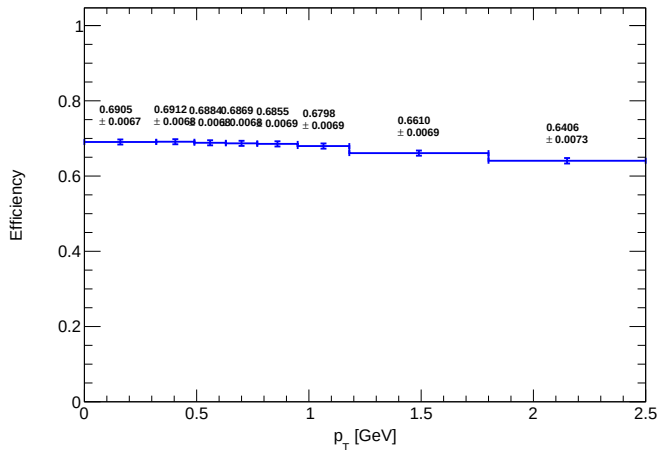
Avg Final Eff (Mix) (LD2)



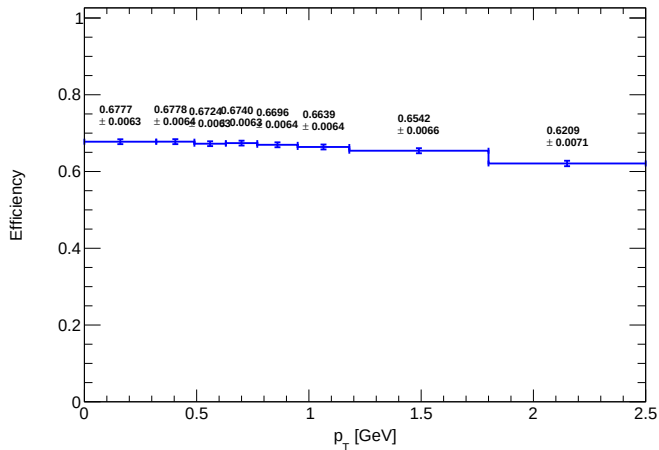
Avg Final Eff (Mix) (Flask)



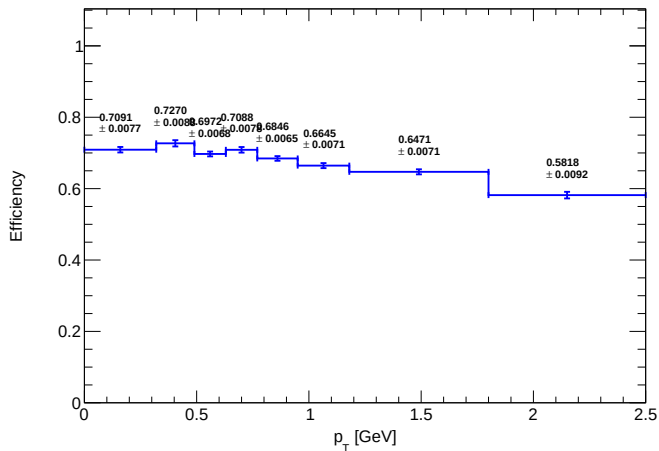
Avg Signal Efficiency (LH2)



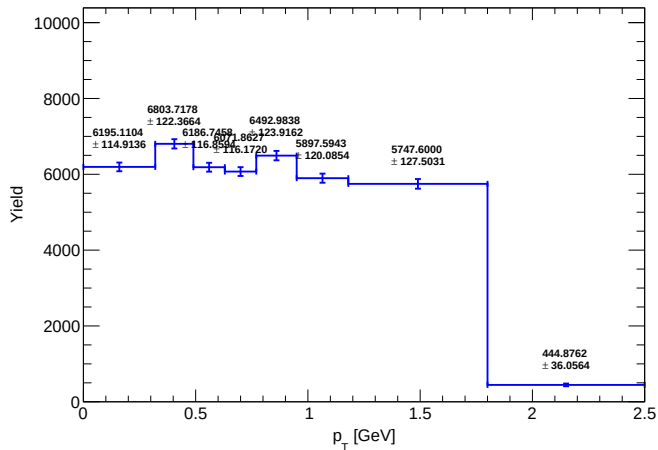
Avg Signal Efficiency (LD2)



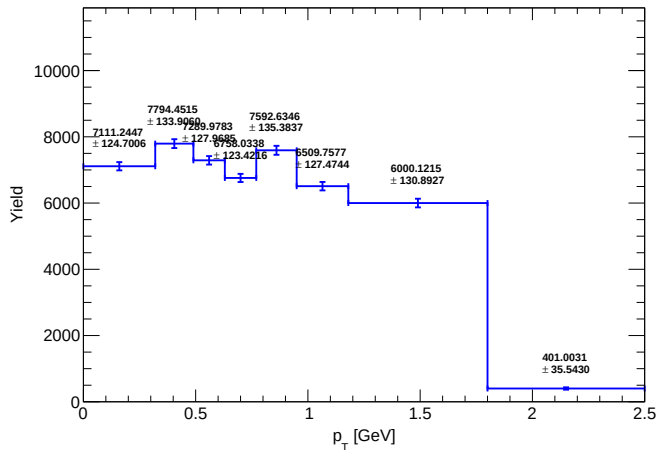
Avg Signal Efficiency (Flask)



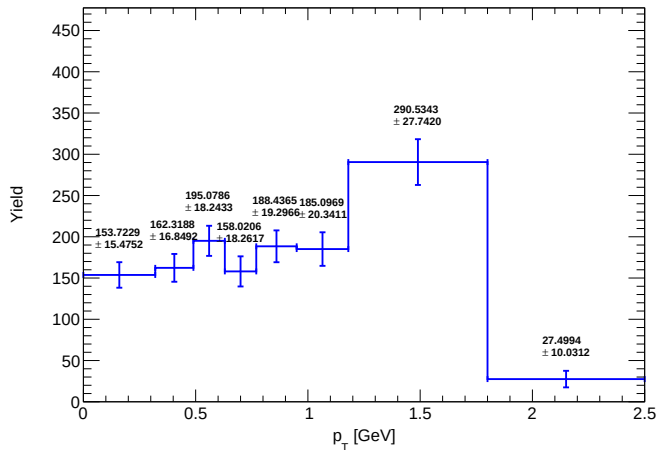
Corrected Yield (Total Error) (LH2)



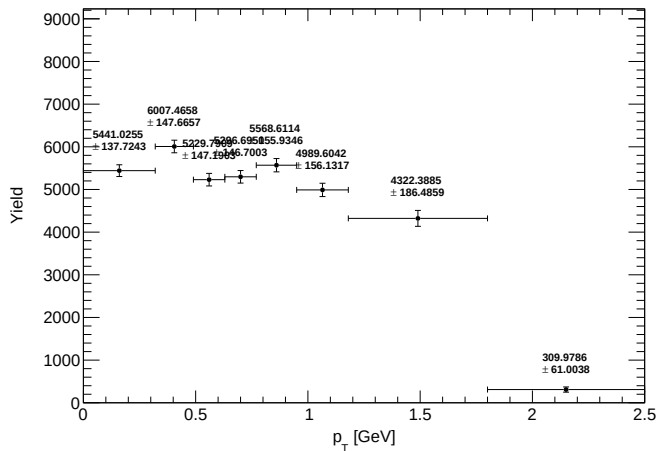
Corrected Yield (Total Error) (LD2)



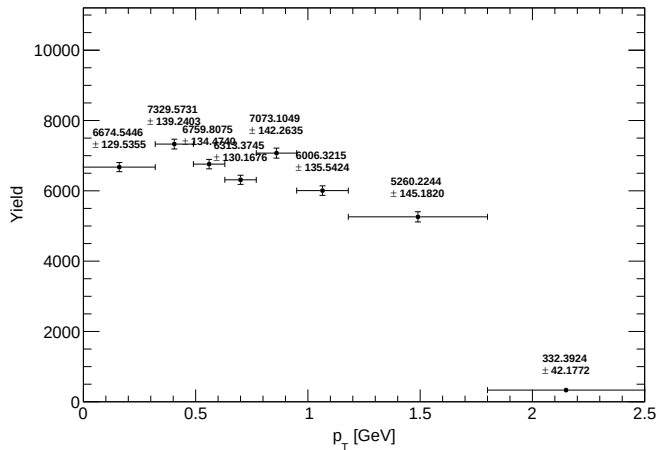
Corrected Yield (Total Error) (Flask)



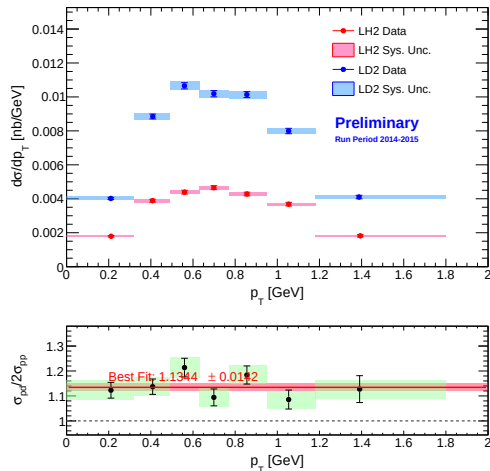
Corrected Yield (LH2 - Flask)



Corrected Yield (LD2 - LH2 - Flask)



Combined XSec Ratio vs p_T true p_T



Implications of introducing cut: $20 < D1 < 385$

Figure 1 Title

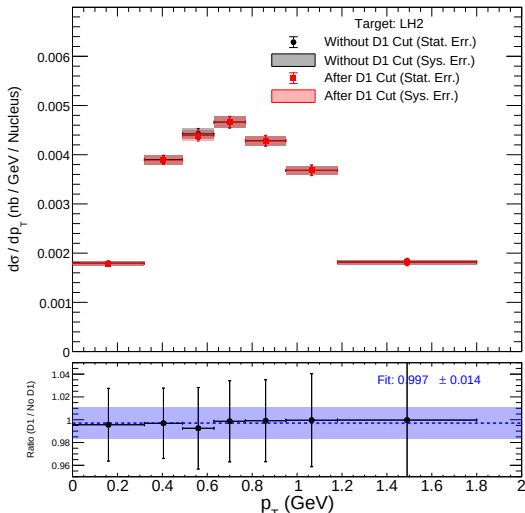


Figure 2 Title

